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Profile:

I am a dedicated interdisciplinary researcher bridging Computer Science and Civil Engineering specializing in AI-driven solutions for sustainable energy, resilient urban systems, and intelligent health monitoring. My doctoral research focused on optimizing Thermal Energy Storage Air Conditioning (TES-AC) systems using Machine Learning to improve energy efficiency in commercial buildings, critical for tropical climates facing global warming. With a strong foundation in Software Engineering and Computer Science, my research interests span renewable energy integration, autonomous vehicle security, AI for climate change mitigation, and multi-agent systems. I also explore AI applications in deep learning, real-time forecasting, smart cities, and digital twin technology. I am committed to advancing AI-driven solutions that address global challenges such as climate resilience, energy transition, and public health. Currently, I am preparing a Marie Skłodowska-Curie Actions proposal and pursuing a European Research Council (ERC) Starting Grant. As a global ambassador for Women Tech Network, I advocate for inclusive AI and sustainable innovation.

Education

Degrees Earned

- Doctor of Philosophy, University of Nottingham (2019–2023)
 - 36-month doctorate (Full-Time) Level of Qualification: 8; Language: English
 - Interdisciplinary: Civil Engineering & Computer Science
 - Research Focus: Applied machine learning for optimizing thermal energy storage air-conditioning (TES-AC) systems in commercial buildings.
 - Adapted to remote collaboration and research methods during the COVID-19 pandemic.
 - Maternity leave 56 weeks, and COVID 19 pandemic mandatory extension 24 weeks.
 - Degree awarded 09th March 2024 (Awarding Institution: University of Nottingham, Administering Institution: University of Nottingham Malaysia)
- Master of Science in Computer Science, University of Nottingham (2018–2019)
 - 12 months Master's (Full-Time); Level of Qualification: 7; Language: English
 - Merit, 64.5%, 180 credits, 90 ECTS.
 - Research Focus: Evaluated virtual reality (VR) as a pedagogical tool for interactive learning. Interdisciplinary project: Human-Computer-Interaction and Civil Engineering Education (Scored 74%)
 - *Subjects:* Compilers; Autonomous Robotic Systems; Linear and Discrete Optimization; Artificial Intelligence Methods; Research Methods; Simulation and Optimisation for Decision Support; Software Engineering Management; Research Project in Computer Science.
 - Degree awarded 22nd February 2020 (Awarding Institution: University of Nottingham, Administering Institution: University of Nottingham Malaysia)
- Bachelor of Science with Honors in Software Engineering, University of Nottingham (2015–2018)
 - 3 years Bachelor's including Mobility (Full-Time) ; Level of Qualification: 6; Language: English
 - Distinction, 72%, 360 credits, 180 ECTS

- Final-Year Project: Developed a GPS-tracking blood donation app to improve emergency medical response. (Scored: 80%)
- *Subjects:*
 - Year 3: Individual Dissertation Single Honours; Software Quality Assurance; Mobile Device Programming; Professional Ethics in Computing; Computer Security; Fundamentals of Information Visualization; Information Visualization Project; Parallel and Distributed Computing.
 - Year 2: Operating Systems and Concurrency; Software Maintenance; Software Engineering Group Project; Algorithms Correctness and Efficiency; Human Computer Interaction; C++ Programming; Languages and Computation; Software Specification.
 - Year 1: Mathematics for Computer Scientists; Introduction to Software Engineering; Computer Fundamentals; Programming and Algorithms; Programing Paradigms; Systems and Architecture; Databases and Interfaces; Fundamentals of Artificial Intelligence.
- Degree awarded 23rd February 2019 (Awarding Institution: University of Nottingham, Administering Institution: University of Nottingham Malaysia)

Research and Teaching Experience

Lecturer (Teaching and Research), Monash University Malaysia (2025-Present)

- Collaborating with interdisciplinary research
- Teaching and Administrative duties for School of Information Technology.

Postdoctoral Researcher (Independent, 2024–2025)

- Involved in interdisciplinary research on AI applications for smart cities, and climate resilience. (Collaboration: Prof Shafi Mohammed Tareq, *Jahangirnagar University*, Assoc. Prof. Jing Ying Wong, *University of Nottingham Malaysia*, Prof. Tomas Maul, *University of Nottingham Malaysia*, EY Open Science Data Challenge 2025).
- Developing machine learning models for early breast cancer diagnosis as part of the ACROBAT Challenge. (Collaboration: Assoc. Prof. Iman Liao, *University of Nottingham Malaysia*, and Cancer Research Malaysia).
- Collaborating with researchers on autonomous vehicle security and AI-driven decision-making. (Collaboration: Prof. Tomas Maul, *University of Nottingham Malaysia*, and Assoc. Prof. Vimal Rau, *University of Nottingham Malaysia*).

Graduate Research & Teaching Assistant, University of Nottingham Malaysia (2019–2023)

- Researched AI-driven optimization of energy efficiency in commercial buildings.
- Researched and designed gamified virtual labs for STEM education (Biology, Chemistry, Stereochemistry, Civil Engineering), integrating innovative technology into higher education. Designed virtual tours for pharmaceutical students and researched soft skill development with CAPE.
- Mentored undergraduate students in computer science.

Teaching Assistant, University of Nottingham Malaysia (2019–2022)

- Supported Software Engineering and Civil Engineering modules.
- Assisted in developing virtual labs for chemistry and biology experiments to enhance interactive learning.

Tutor, and Invigilator, IGCSE (Edexcel and Cambridge, British Council), (2013-2015)

- Personal tutor for primary and secondary students.
- Engaged in IGCSE course tuitions and invigilation.

Key Research Interests

- AI for Sustainable and Resilient Cities for Climate Action and Energy Security
- Human-Centric AI & Decision-Making Models
- Autonomous Vehicle Security & V2X Communication
- AI in Higher Education & Gamified Learning

Potential Referees

1. Prof. Tomas Maul- Co Supervisor and Academic mentor
Professor, University of Nottingham Malaysia (Computer Science)
Email: Tomas.Maul@nottingham.edu.my
2. Dr. Jing Ying Wong-Main Supervisor
Associate Professor, University of Nottingham Malaysia (Civil Engineering)
Email: JingYing.Wong@nottingham.edu.my
3. Dr. Iman Yi Liao- Undergraduate and Postgraduate module convenor and co-researcher
Associate Professor, University of Nottingham Malaysia (Computer Science)
Email: Iman.Liao@nottingham.edu.my
4. Dr. Chun-Chieh Yip- Co Supervisor. Head of Department, University Tunku Abdul Rahman (Civil Engineering) Email: yipcc@utar.edu.my

Publications

- Charging Water Load Prediction for a Thermal-Energy-Storage Air-Conditioner (*Journal of Building Engineering*, Elsevier) – doi:10.1016/j.jobee.2023.107016
- Effects of External Weather on the Water Consumption of Thermal-Energy-Storage AC (*Energy Nexus*, Elsevier) – doi:10.1016/j.nexus.2023.100187
- Application of deep learning in facility management and maintenance for heating, ventilation, and air conditioning (*Automation in Construction*, Elsevier) – doi:10.1016/j.autcon.2022.104445
- Lecture-based, Virtual Reality Game-based and Their Combination: Which is Better for Higher Education?- doi:10.1108/JARHE-09-2020-0302
- Gamified Virtual Labs: Shifting from physical environments for low-risk interactive learning (*Journal of Applied Research in Higher Education*, Emerald Insight) – doi:10.1108/JARHE-09-2022-0281
- Personnel Training for Common Facility Management Issues in Thermal-Energy-Storage Chiller Plant Using a Serious 3D Game – doi:10.1177/10468781241232594
- Chapter 6: The Potential of Deep Learning in Dynamic Maintenance Scheduling for Thermal Energy Storage Chiller Plants, *Book: Interpretable Machine Learning for the Analysis, Design, Assessment, and Informed Decision Making for Civil Infrastructure*- doi:10.1016/B978-0-12-824073-1.00012-5

Grants and Recognitions

- Science and Engineering Postgraduate Research Scholarship for PhD (2019-2023)
- WomenTech Global Awards – Second Place IT Graduate of the Year (2022)
- AWS AI & ML Advanced Program Scholarship (2023)
- WomenTech Speaker
- Nominated for Women in AI APAC (Research and Education)
- Nominated for Women of the Future ESG, 50 Rising Stars (Planet) (2025)